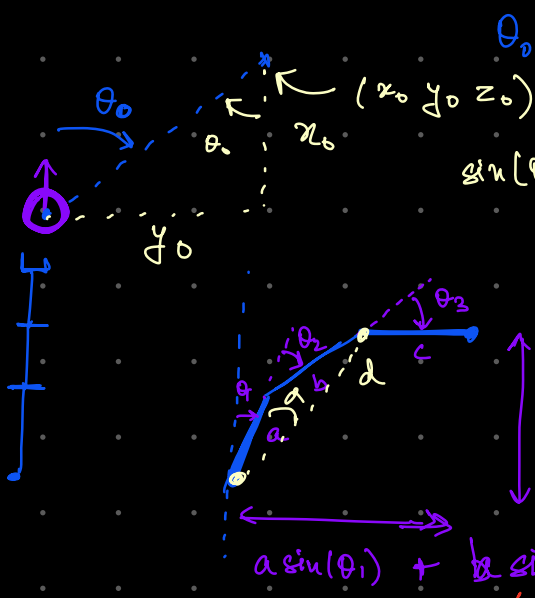


(EE)
 wrist
 elbow
 shoulder



$\theta_0 \rightarrow$ shoulder pan

$$\sin(\theta_0) = \frac{y_0}{\sqrt{x_0^2 + y_0^2}} \quad \left\{ \begin{array}{l} \text{Not Sufficient} \\ \text{Since we need} \\ 0 \text{ to } 2\pi \end{array} \right.$$

* Add an "if cond"

$$a \cos(\theta_1) + b \cos(\theta_1 + \theta_2) + c \cos(\theta_1 + \theta_2 + \theta_3) = z_0$$

$$a \sin(\theta_1) + b \sin(\theta_1 + \theta_2) + c \sin(\theta_1 + \theta_2 + \theta_3) = \sqrt{x_0^2 + y_0^2}$$

one additional dof
gripper Angle (θ_3) = 0

Triangle
cosine rule
for θ_2

$$\theta_2 = \cos^{-1}\left(\frac{a^2 + b^2 - d^2}{2ab}\right), \quad \theta_1 = \pi - (\alpha) - \theta_2$$